VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Module Bank

Module 2

Academic Year 2025-26

Staff Name: Dr. Sreekar Guddeti

Program Name: B. Tech

Branch: ECE + ECEVLSI

Year: 1

Semester: 2

Course: EP

Code: 25PY101

Section number: 24

Instructions:

1. Answer all questions.

1. a) Write the postulates of the Quantum free electron theory.

   b) What is Fermi-Dirac distribution function and explain its variation of the Fermi-Dirac distribution function at T=0K and T>0K.

   c) Calculate the temperature at which there is 3% probability of finding the electron having energy more than 1eV above the Fermi level?

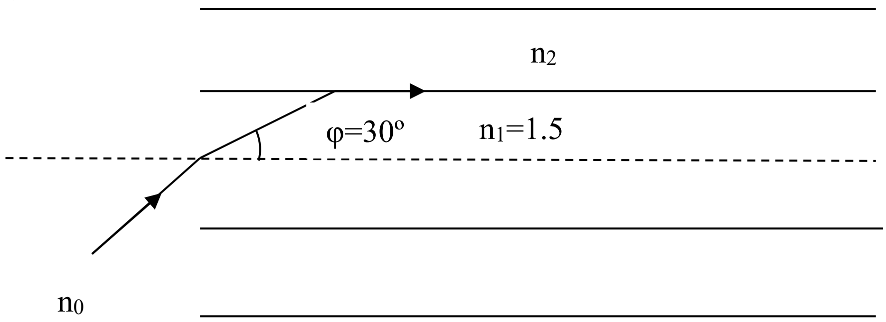
2. a) What is the principle of a solar cell? How the light energy is converted into electrical energy using a solar cell?

   b) Draw the I-V characteristics and obtain an expression for the Efficiency and Fill factor.

   c) A solar cell is having an area of 1cm ×1cm and the input light power density is 1000W/m^2. The efficiency of the solar cell is 5%. Calculate the load resistance value to withdraw a maximum output power when a current of 5 mA is flowing in the circuit.

3. a) How the numerical aperture and acceptance angle are related with each other and draw the mathematical relation between them

   b)



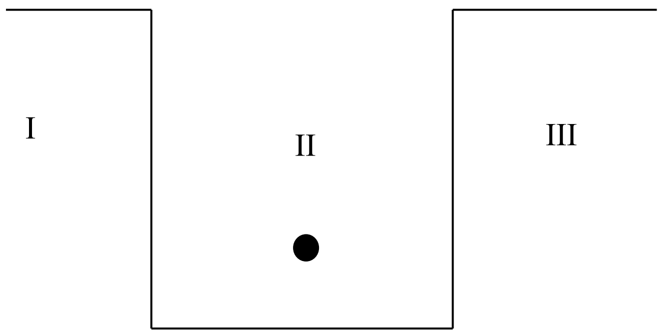
The light ray is launching from the air to the core medium which has a refractive index of 1.5 and the angle refraction is 30° as depicted in the above figure. Calculate

i) Refractive index of the cladding

ii) Maximum Launching angle or Acceptance angle

c) Also calculate the numerical aperture and fractional refractive index of the optical fiber.

4. a) An electron is confined between the two boundaries of height V_o as shown in below figure. The energy of the particle is less than the height of the barrier ($E<V_o$). Apply the Schrodinger equation in the regions I and III and find the probability.



b) Explain the concept of tunnelling by plotting the wavefunctions of the finite potential well

c) Compare the wavefunctions of the finite and infinite potential wells.

5. a) Write a note on Heisenberg uncertainty principle and its significance.

b) Prove that a free electron cannot exist inside a nucleus using the Heisenberg uncertainty principle.

c) Electrons moving with a velocity of 3.32×10^5 m/s if this velocity is measured with an inaccuracy of 1% then estimate the uncertainty in the position of an electron.

6. a) Outline the characteristic features of lasers.

b) Analyse population inversion and various pumping mechanisms in lasers.

c) The wavelength of light emitted from a laser is 610 nm at room temperature. Determine the ratio of the population of higher energy state to lower energy state. How the ratio of the population is effected, if the temperature is changed from room temperature to 400 K?

7. a) What is meant by molar-specific heat?

b) Explain how the classical free electron theory is failed to explain the electronic specific heat.

c) Elucidate how Quantum free electron theory is successful in explaining electronic-specific heat by overcoming the failures of the classical free electron theory.

8. a) What is meant by stimulated emission. Explain the necessity of stimulated emission in obtaining the properties of LASER.

b) Explicate the essential conditions to obtain a LASER?

c) Elucidate the working of semiconductor LASER with the help of a band diagram.

9. a) Starting from the expressions of electron and hole concentrations, obtain an expression for the electrical conductivity of an intrinsic semiconductor.

b) Why Germanium has more conductivity than Gallium Arsenide at all temperatures?

c) The conductivity of Si at 20 °C is 4.4×10^{-4} $(\Omega m)^{-1}$. Calculate its conductivity at 100 °C given the bandgap of Si is 1.1eV.

10. a) Both solar cell and photodiode convert the light energy into electrical energy. What are the basic differences between them in terms of operation?

b) Describe the working mechanism of photodiode.

c) Draw the I-V characteristics of the photodiode as a function of incident light intensity.

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25PY101
Engineering Physics

Year 1 Semester 2 Section 24 (ECE + ECEVLSI)

Course instructor: Dr. Sreekar Guddeti

May 9, 2026

Module Bank

Module 2

Instructions:

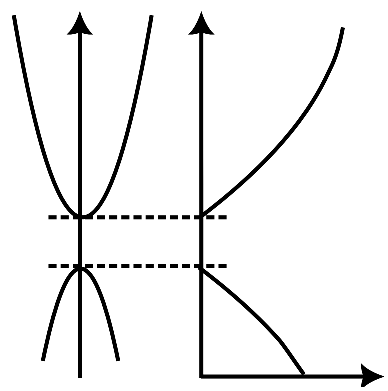
1. Answer all questions.
2. BT stands for Blooms Taxonomy.

PART-B

(1) Fermi level in semiconductors [10 M]

(a) Below is the partially drawn energy band structure of an intrinsic semiconductor. Clearly label the following in the illustration. [BT 1][3 M]

(i) Valence band
(ii) Conduction band
(iii) Conduction band minimum
(iv) Valence band maximum
(v) Conduction band density of states
(vi) Valence band density of states



(b) Calculate position of Fermi level with respect to center of the bandgap in silicon for [BT 2][2 M]

(i) $T = 0\,\mathrm{K}$
(ii) $T = 300\,\mathrm{K}$

The electron effective mass is $1.08m_0$ and hole effective mass is $0.56m_0$.

(c) Draw, compare, and contrast the sketches of the energy band diagram for p-type semiconductor at $T = 0\,\mathrm{K}$ and $T = 300\,\mathrm{K}$. [BT 3][2 M]

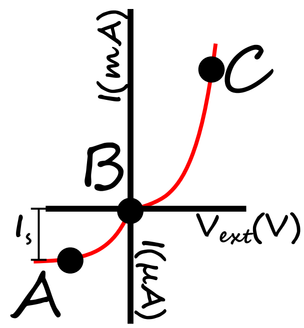
(d) Silicon atoms, at a concentration of $7 \times 10^{15}\,\mathrm{cm}^{-3}$, are added to GaAs. Assume that the silicon atoms act as fully ionized dopant atoms and that 5% of the dopants replace Ga atoms and 95% replace As atoms. [BT 2][3 M]

(i) Determine the donor and acceptor concentrations.
(ii) Is the material n type or p type?

(2) p-n junction diode [10 M]

(a) Below is the sketch of the $I - V$ characteristics of pn junction diode. Three operating points O_A, O_B, O_C, have been identified. Fill the table of operating points. [BT 1][3 M]

Operating point	Operating bias	Applications	Working principle
O_A			
O_B			
O_C			



Operating points of pn-junction diodea

(b) Define reverse saturation current of pn junction diode. [BT 1][1 M]

(c) A silicon solar cell under standard test conditions has short circuit current density of 35 mA cm^{-2}, and open circuit voltage of 0.60 V. The fill factor for the material is 0.80. If the incident power is 100 mW cm^{-2}, calculate the efficiency of the cell. [BT 3][4 M]

(d) The energy band gap of a semiconductor is 3 eV. Is this material useful for solar cell applications? Give reasons. [BT 3][2 M]

(3) Direct and indirect band gap semiconductors [10 M]

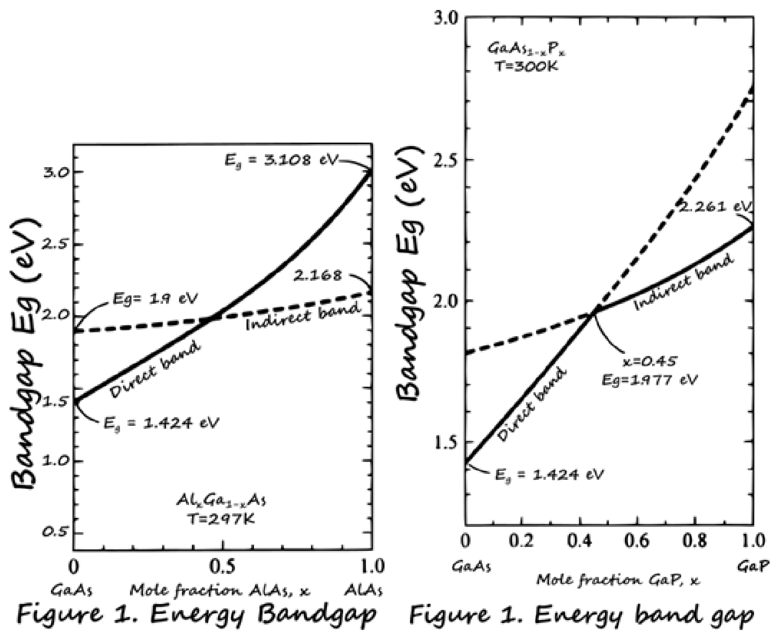
(a) Sketch the E vs k diagram for [BT 1][2 M]

(i) a direct bandgap semiconductor, and

(ii) an indirect bandgap semiconductor.

(b) Why is direct band gap semiconductor useful for solar cell application? [BT 2][2 M]

(c) $Al_xGa_{1-x}As$ is a ternary semiconductor and the energy band gap variation as a function of mole fraction x of AlAs is shown in Figure. 1a. Which composition can used as material for blue LED application? [BT 3][3 M]



(a) (b)

Figure 1: Band gap variations in ternary semiconductors (a) $Al_xGa_{1-x}As$, and (b) $GaAs_{1-x}P_x$.

(d) The energy band gap variation as a function of mole fraction x of $GaAs_{1-x}P_x$ is shown in Figure. 1b. What is the range of colours that can be emitted if this material is used for LED application? Report in wavelength. [BT 3][3 M]

Property of photon	Range	
	Maximum	Minimum
Energy Wavelength		